

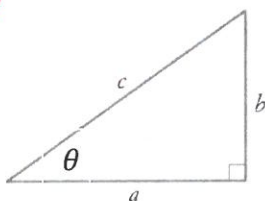
|                                                                                   |                                                                                                           |  |                           |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--|---------------------------|
| Name:                                                                             | <u>SOLE UTIONS</u>                                                                                        |  | Date: _____               |
| Teacher:                                                                          |                                                                                                           |  |                           |
|  | <b>Year 9 Mathematics</b><br><b>Topic – Trigonometry</b>                                                  |  | <b>SCORE:</b><br><br>/ 35 |
|                                                                                   | <u>Working out MUST BE SHOWN for full marks</u>                                                           |  |                           |
| Total Time:                                                                       | 45 minutes                                                                                                |  |                           |
| Weighting:                                                                        | 6 %                                                                                                       |  |                           |
| Equipment:                                                                        | To be provided by the student:<br>Pens and Pencils, scientific calculator, ruler and an A4 page of notes. |  |                           |

### Multiple choice questions

6 marks

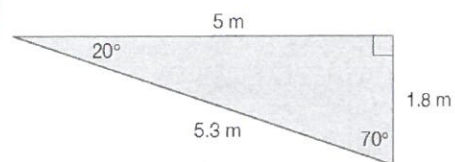
**Question 1.** Which statement is true for the angle  $\theta$  in the diagram?

- A.  $b = \text{opposite}$  ✓
- B.  $b = \text{adjacent}$
- C.  $c = \text{adjacent}$
- D.  $a = \text{opposite}$



**Question 2.** What is the length of the adjacent side when considering the  $20^\circ$  angle?

- A. 20 m
- B. 5 m ✓
- C. 1.8 m
- D. 5.3 m

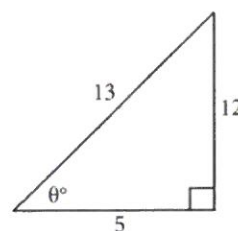


**Question 3.** What is the ratio for Sine?

- A.  $\frac{HYP}{OPP}$
- B.  $\frac{OPP}{ADJ}$
- C.  $\frac{OPP}{HYP}$  ✓
- D.  $\frac{ADJ}{OPP}$

**Question 4.** Which is the correct ratio for  $\tan \theta$

- A.  $\tan \theta = \frac{13}{12}$
- B.  $\tan \theta = \frac{5}{12}$
- C.  $\tan \theta = \frac{5}{13}$
- D.  $\tan \theta = \frac{12}{5}$  ✓

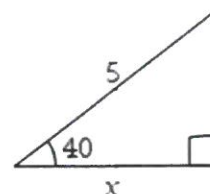


**Question 5.** Which of the following is a Pythagorean triple?

- A. (1, 2, 3)
- B. (4, 5, 6)
- C. (5, 12, 13) ✓
- D. (15, 20, 30)

**Question 6.** Which ratio would you use to determine the value of  $x$ ?

- A. sine
- B. cosine ✓
- C. tangent
- D.  $\frac{x}{5} = 40$



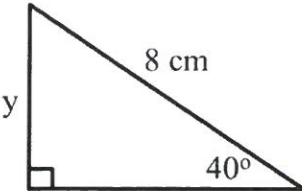
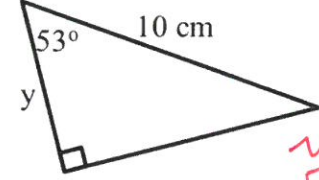
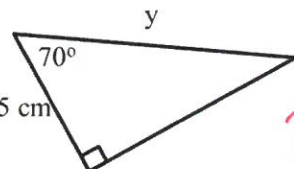
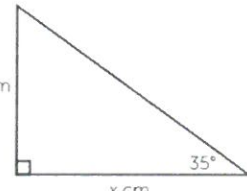
6

# Short answer questions

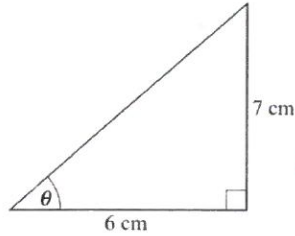
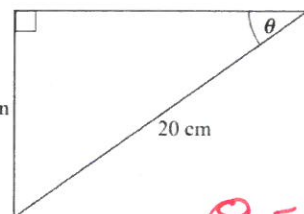
29 marks

Show FULL working out to get FULL marks

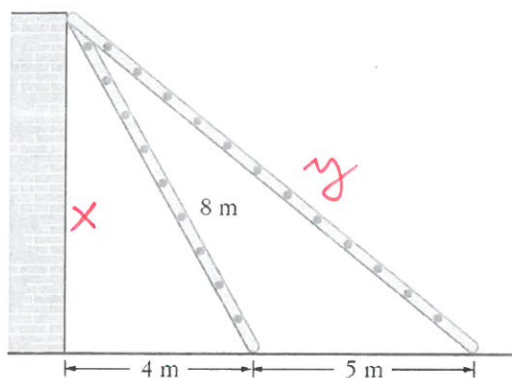
**Question 7.** Calculate the lengths of the sides marked with letters in each of the following triangles, giving your answers to 1 decimal place. [8 marks]

|                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a.</p>  <p><math>\sin \theta = \frac{O}{H}</math> ✓<br/> <math>y = 8 \times \sin 40^\circ</math><br/> <math>= 5.1 \text{ cm}</math> ✓</p>    | <p>b.</p>  <p><math>\cos \theta = \frac{A}{H}</math> ✓<br/> <math>y = 10 \times \cos 53^\circ</math><br/> <math>= 6.0 \text{ cm}</math> ✓</p>   |
| <p>c.</p>  <p><math>\cos \theta = \frac{A}{H}</math><br/> <math>y = \frac{5}{\cos 70^\circ}</math> ✓<br/> <math>= 14.6 \text{ cm}</math> ✓</p> | <p>d.</p>  <p><math>\tan \theta = \frac{O}{A}</math><br/> <math>x = \frac{8}{\tan 35^\circ}</math> ✓<br/> <math>= 11.4 \text{ cm}</math> ✓</p> |

**Question 8.** Calculate the size of each of the angles marked  $\theta$  in the following triangles, giving your answers to the nearest degree. [4 marks]

|                                                                                                                                                                                                                              |                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p>  <p><math>\tan \theta = \frac{O}{A}</math> ✓<br/> <math>\theta = \tan^{-1}(7 \div 6)</math><br/> <math>= 49^\circ</math> ✓</p> | <p>(b)</p>  <p><math>\sin \theta = \frac{O}{H}</math> ✓<br/> <math>\theta = \sin^{-1}(15 \div 20)</math><br/> <math>= 49^\circ</math> ✓</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Question 9.** Two ladders are resting at the top of a wall. The shorter ladder is 8m long and its bottom end is 4m from the base of the wall. The end of the longer ladder is 5m from the end of the shorter ladder on the ground. How long is the longer ladder to the nearest metre? [5 marks]



$$1 \text{ --- } 9\text{m} \text{ --- } 1$$

✓

$$a^2 = c^2 - b^2$$

$$x = \sqrt{8^2 - 4^2}$$

$$= \sqrt{48}$$

$$\approx 6.93\text{m}$$

✓

$$c^2 = a^2 + b^2$$

✓

$$c = \sqrt{9^2 + 48}$$

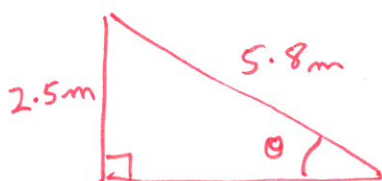
$$= 11.4$$

✓

LADDER IS 11m LONG ✓

**Question 10.** A children's slide has a length of 5.8m and makes an angle of  $38^\circ$  with the vertical ladder that is 2.5m high. [3 marks]

a) Draw a diagram showing the information provided



b) Find the angle the slide makes with the ground.

$$\sin \theta = \frac{o}{h}$$

✓

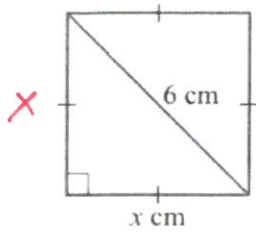
$$\theta = \sin^{-1}(2.5 \div 5.8)$$

$$= 25.5^\circ$$

✓

8

**Question 11.** A square has diagonals of length 6 cm. Find the length of its sides, correct to 2 decimal places. [3 Marks]



$$\begin{aligned}
 C^2 &= x^2 + x^2 \\
 6^2 &= 2x^2 \\
 \frac{36}{2} &= x^2 \\
 x &= \sqrt{18} \\
 &= 4.24 \text{ cm}
 \end{aligned}$$

**Question 12.** The following answers were given on a trigonometry test. [6 marks]

|                                                                                                                                                                                                                   |                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(i) Find the value of <math>k</math>.</p> $  \begin{aligned}  \cos \theta &= \frac{A}{H} \\  \cos 40^\circ &= \frac{k}{8} \\  8 \times \cos 40^\circ &= k \\  k &= 6.13 \text{ m}  \end{aligned}  $            | <p>(ii) Find the value of <math>x</math>.</p> $  \begin{aligned}  \sin \theta &= \frac{O}{H} \\  \sin 55^\circ &= \frac{13}{x} \\  13 \times \sin 55^\circ &= x \\  x &= 10.65 \text{ km}  \end{aligned}  $               |
| <p>a. What mistake did this student make?</p> <p style="color: red; text-align: center;">WRONG ANGLE<br/>SHOULD BE <math>50^\circ</math> ✓</p>                                                                    | <p>a. What mistake did this student make?</p> <p style="color: red; text-align: center;">CROSS MULTIPLICATION<br/>INCORRECT ✓</p>                                                                                         |
| <p>b. Show the correct calculations and answers for the question above</p> $  \begin{aligned}  \cos \theta &= \frac{A}{H} \\  k &= 8 \times \cos 50^\circ \\  &= 5.1 \text{ m} \quad \checkmark  \end{aligned}  $ | <p>b. Show the correct calculations and answers for the question above</p> $  \begin{aligned}  \sin 55^\circ &= \frac{13}{x} \\  x &= \frac{13}{\sin 55^\circ} \\  &= 15.9 \text{ km.} \quad \checkmark  \end{aligned}  $ |